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Randomized Control Trials

Early time-restricted eating with energy restriction has a better effect on body fat mass, diastolic blood pressure, metabolic age and fasting glucose compared to late time-restricted eating with energy restriction and/or energy restriction alone: A 3-month randomized clinical trial



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SUMMARY

Background & aims: Time-restricted eating (TRE) has attracted increasing attention from researchers and the public. Recent studies suggest that the combination of TRE with energy restriction (ER) may have more favourable effects on both physical and biochemical aspects compared to ER alone. The aim of the present 3-month intervention study was to determine the effects of an 8-h early time-restricted eating and an 8-h late time-restricted eating with ER (eTRE + ER and lTRE + ER) compared to 12-h ER alone on body mass and other anthropometric and cardiometabolic risk factors in participants with overweight and obesity.

Methods: Participants ($n = 108$) were allocated to three different groups according to their personal chronotype: eTRE + ER (37 participants), lTRE + ER (37 participants) and ER (34 participants). Ninety-three participants completed the entire 3-month intervention (34 in eTRE + ER, 28 in lTRE + ER and 31 in ER). Anthropometric and cardiometabolic risk factors were measured at baseline and after 1, 2 and 3 months of the intervention. Sleep quality and quality of life were assessed at baseline and after 3 months of the intervention. ER was determined based on the individual's resting metabolic rate. Effects were analyzed using the *per-protocol* approach.

Results: Results showed a significant time main effect ($p < 0.001$), suggesting a decrease in body mass at the end of the 3-month intervention with a mean loss of -5.0 kg (95 % CI, $-5.7, -4.3$) for the eTRE + ER group, -4.4 kg (95 % CI, $-5.2, -3.6$) for the lTRE + ER group and -4.3 kg (95 % CI, $-5.0, -3.6$) for the ER group, with no significant difference between the groups ($p = 0.319$). eTRE + ER had greater improvements in fat mass (-1.2 % (95 % CI, $-2.1, -0.2$), $p = 0.013$) and fasting glucose (-0.35 mmol/L (95 % CI, $-0.63, -0.06$), $p = 0.012$) than participants in the lTRE + ER group and greater improvements in fat mass (-1.1 % (95 % CI, $-2.0, -0.1$), $p = 0.022$), metabolic age (-3 years (95 % CI, $-5, -0$), $p = 0.028$) and diastolic blood pressure (-4 mmHg (95 % CI, $-8, -0$), $p = 0.033$) than the participants in the ER group. No significant changes were found between the groups for the other parameters measured.

Conclusions: There was no difference in body mass between the eTRE + ER, lTRE + ER and ER groups after 3 months of intervention. However, eTRE + ER showed a greater benefit for fasting blood glucose, certain anthropometric parameters and diastolic blood pressure compared to lTRE + ER and/or ER alone. Other anthropometric, biochemical and health-related parameters were not affected by eating window.

Trial Registration: <https://clinicaltrials.gov/study/NCT05730231>.

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1. Introduction

Eating behaviour in modern society is characterized by prolonged, energy-rich and unbalanced daily eating patterns, which, together with a Western-style diet, a sedentary lifestyle and chronic sleep deprivation, contribute to the development of obesity [1]. Obesity is considered a multifactorial pathology and a low-grade chronic inflammatory disease. It correlates with the occurrence of metabolic syndrome (MetS), which includes insulin resistance, visceral obesity, dyslipidaemia and hypertension [2], and is also associated with a lower quality of life and poorer quality of sleep [3–5].

Optimal diets and effective weight management strategies are still limited and are a topic of discussion among researchers and healthcare professionals. According to a meta-analysis of different diet programmes, energy restriction (ER) is the primary driver of weight loss, regardless of the diet followed [6]. ‘Time-restricted eating’ (TRE) has gained increasing popularity in recent years and is a potential new chrono-nutritional strategy for weight loss and improvement of metabolic health due to the shorter eating time and thus easier daily adherence to the dietary pattern. To date, many of the intervention studies on TRE have largely ignored the type of food consumed, its quantity, macronutrient and energy distribution across meals, and the number of meals consumed during the eating window [7]. Nevertheless, some studies have shown that TRE results in spontaneous reduction in daily energy intake (20–30 %) [8,9]. TRE has been shown to have many beneficial metabolic effects in humans [8–14], as well as improved quality of life and sleep quality [15,16]. Some of the above effects may also depend on the particular time window [10,11,17]. For example, a recent study has shown that skipping breakfast increases the risk of overweight and obesity [18] and that eating late in the evening is associated with obesity and MetS [19–21]. An interesting but still unanswered question is whether the health benefits of TRE result from the contributions of ER, TRE or the alignment of meal timing with circadian oscillations.

A previous study has shown that the timing of food intake, tailored to an individual’s chronotype is important for the success and maintenance of dietary strategies [22]. Therefore, in our study, we matched participants as closely as possible to their individual chronotype. In overweight and obese participants with at least one additional MetS component, we assessed anthropometric and cardiometabolic risk factors, sleep quality and quality of life in the early TRE (eTRE) + ER group (eating between 8:00–16:00) for the morning and intermediate chronotypes, in the late TRE (lTRE) + ER group (eating between 12:00 and 20:00) for the evening and intermediate chronotypes, and compared them with the ER group (eating between 8:00 and 20:00) for the morning, evening and intermediate chronotypes.

Importantly, our participants had a prescribed dietary plan with the same daily distribution of macronutrients and energy. In the present study, we hypothesized that TRE + ER (both eTRE + ER and lTRE + ER) would be more effective in weight loss and improving some cardiometabolic risk factors than ER alone.

2. Methods

2.1. Study design and study participants

The study was a three-arm, parallel, 3-month interventional study conducted at the University of Primorska, Faculty of Health Sciences, between March and June 2023. The study protocol was approved by the Slovenian National Medical Ethics Committee (No. 0120–557/2017/4; Ministry of Health, Republic of Slovenia) and registered at ClinicalTrials.gov (NCT05730231).

Participants were recruited via online, radio and public television advertisements, social media and local newspapers. Interested volunteers signed up via an online entry questionnaire designed by the researchers, which included questions on personal and demographic data, lifestyle habits, health status and some anthropometric parameters (current body mass, body height, waist circumference and blood pressure). The completed questionnaires were checked by the researchers for fulfilment of the inclusion and exclusion criteria.

The inclusion criteria were as follows: participants aged 18–60 years with a body mass index (BMI) between 25 kg/m² and 35 kg/m² and with one of the following MetS parameters [23]: increased waist circumference ≥94 cm in men and ≥80 cm in women; fasting glucose ≥5.6 mmol/L; fasting plasma triacylglycerols ≥1.7 mmol/L; high-density lipoprotein (HDL) cholesterol ≤ 1.0 mmol/L in men and ≤1.3 mmol/L in women; systolic blood pressure ≥130 mmHg and diastolic blood pressure ≥85 mmHg.

The exclusion criteria were: 1) taking medications for hypertension or hypercholesterolaemia, 2) average daily eating window <11 h/day, 3) pregnancy or lactation, 4) smoking, 5) participation in a weight-loss program or unstable body mass in the last three months before the start of the study (>5 % weight loss), 6) chronic health conditions (e.g., cardiovascular diseases (CVD), type 1 or type 2 diabetes mellitus (T2DM), 7) gastrointestinal, psychiatric, rheumatological, oncological, haematological or endocrine diseases, 8) diagnosis of an eating disorder or history of an eating disorder, 9) shift work, 10) alcohol use of >2 servings/day for men and >1 serving/day for women, 11) use of food supplements that could affect the main outcomes of the study (e.g. cholesterol-lowering drops, vitamin C, coenzyme Q10, collagen). All participants received detailed information about the purpose and procedure of the study and signed a written informed consent form.

2.2. Intervention

Before the start of the 3-month intervention study, the participants were allocated to one of the following three groups: eTRE + ER (eating window between 8:00 and 16:00), lTRE + ER (eating window between 12:00 and 20:00) and ER group (eating window between 8:00 and 20:00). The allocation was based on the Morningness-Eveningness Questionnaire (MEQ) [24].

The MEQ can define individual differences in the sleep–wake cycle and the time of the day when you feel and perform best. The questionnaire consists of 19 items and is aimed at measuring several aspects of morningness, including sleep habits, sleepiness and preferred time for activities that require physical or mental alertness. The response options include a mixture of multiple-choice options and a 4–5 point ordinal scale to indicate preferred time for activities. The score on this questionnaire can range from 16 to 86 and classifies individuals into extreme evening type (16–46 points), evening type (47–52 points), intermediate type (53–64 points), morning type (65–69 points) and extreme morning type (70–86 points) [24]. In the present study, participants with a total MEQ score >58 were considered as extreme morning, morning or intermediate type and randomly assigned in a 2:1 ratio to either the eTRE + ER or ER group. Similarly, participants with an MEQ score ≤58 were considered extreme evening, evening or intermediate type and randomly assigned to either the lTRE + ER or ER group, again in a 2:1 ratio. After randomisation, the median MEQ score was 64 in the eTRE + ER group, 54 in the lTRE + ER group and 58 in the ER group. Randomization with stratification was performed using free open-source desktop application, MinimPy (<https://sourceforge.net/projects/minimpy>) [25]. The stratified variables were sex, age (18–39 and 40–60 years) and BMI (25.0–29.9 kg/m² and 30.0–34.9 kg/m²). The randomization was

done before the baseline measurement by a research assistant who did not know the aim of the study.

The 3-month intervention study consisted of four measurements and three motivational sessions (Fig. 1). At baseline, after the first, second and third month of the intervention, the measurements included anthropometric measurements, blood pressure, and biochemical measurements. Resting metabolic rate was measured at the beginning and after three months of the intervention. The measurements were carried out in the morning; scheduling was based on group allocation, so that fasting lasted 13–15 h for each participant. Participants were also instructed to drink only water and not to consume caffeine in the morning, and avoid vigorous physical activity on the day before the measurements.

Moreover, all participants received nutritional counselling at the beginning of the study. During this visit, a dietitian calculated the participants' daily energy requirements and prepared a dietary plan for the 3-month intervention. The daily energy requirement of each participant was calculated based on the individual resting metabolic rate multiplied by the self-reported physical activity level (PAL) and reduced by 2093 kJ/500 kcal. If the prescribed daily energy intake was lower than the participant's resting metabolic rate, the ER was adjusted to ensure the individual's daily energy requirements were met. PAL and physical activity levels were assessed using the International Physical Activity Questionnaire (IPAQ) short form. The IPAQ short form asks about three specific

types of activity and sitting. The specific types of activity assessed are walking, moderate–intensity activities and vigorous intensity activities; frequency (measured in days per week) and duration (time per day) are collected separately for each specific type. Computation of the total score requires summation of the duration (in minutes) and frequency (days) of walking, moderate- and vigorous–intensity activity. The IPAQ categorisation is moderate, low and high physical activity levels [26].

All participants were prescribed a dietary plan based on their daily energy requirements to follow on their own. Dietary plan were the same for all participants and were formulated to supply 45–50 % energy by carbohydrates, 15–20 % by proteins and 30–35 % by fats. The daily energy intake was evenly distributed over three meals: 30 % of the daily energy intake at breakfast, 40 % at lunch and 30 % at dinner. Snacks between meals were not allowed. Outside the eating window, only water and tea without added sugar were allowed. Participants were instructed to eat only the meals prescribed by the dietitian and at the time indicated.

To improve adherence to the intervention protocols, the qualified dietitian provided participants with a detailed 10-day menu plan based on their energy requirements, with all food ingredients expressed by weight and/or volume, and a handout containing details of the assigned diet with possible substitutions. In addition, all participants had monthly motivational meetings with a dietitian, which took place in three groups: eTRE + ER, ITRE + ER and ER. The aim of the monthly meetings was to educate the participants

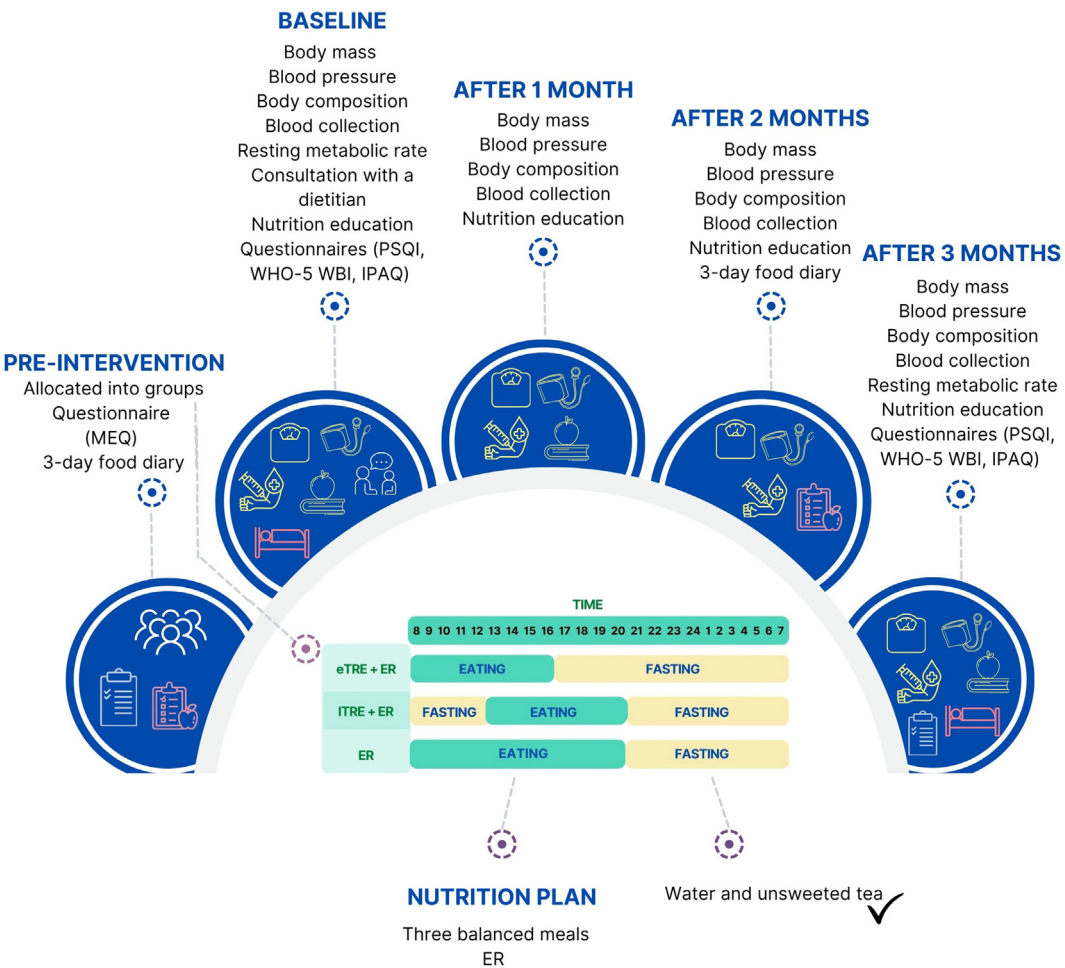


Fig. 1. Study protocol. ER, energy restriction; eTRE, early time-restricted eating; IPAQ, International Physical Activity Questionnaire; ITRE, late time-restricted eating; MEQ, Morningness-eveningness questionnaire; PSQI, Pittsburgh sleep quality index; WHO-5 WBI, WHO-5 Well-being Index.

about general nutritional information. The motivational sessions consisted of the following main topics: energy values of food, nutrition facts labels, food preparation, and emotional eating. In addition, the sessions served also as an opportunity to ask questions or clarify doubts about the intervention, report adverse effects and encourage participants to follow all instructions during the study and measurements.

2.3. Outcome measures

The primary outcome was the mean change in body mass between eTRE + ER, ITRE + ER and ER from baseline to the end of the 3-month intervention. The secondary outcomes were mean changes in body composition (percentage of body fat, visceral fat rating, fat free mass, muscle mass, waist circumference and metabolic age), resting metabolic rate, and fasting cardiometabolic risk factors (blood pressure, glucose, lipid profile and C-reactive protein (CRP) between eTRE + ER, ITRE + ER and ER from baseline to the end of the intervention. Moreover, we also assessed quality of life and sleep quality as tertiary outcomes. Quality of life, sleep quality and resting metabolic rate were assessed at baseline and after 3-month of intervention, whereas all other outcomes at baseline, after 1, 2, and 3 months of intervention.

2.4. Body mass, body composition and blood pressure

Body height was measured using a standard stadiometer to the nearest 0.5 cm. Anthropometric parameters (fat mass (%), visceral fat rating, fat free mass, muscle mass, metabolic age) were measured in light clothing without shoes using Tanita MC-980MA bioelectrical impedance analyser (BIA) (Maeno-cho, Japan) and dedicated software (GMON Pro-Tanita). Body mass was measured to the nearest 0.1 kg. BMI was calculated as a quotient of body mass (kg) and height (m) squared. Waist circumference was measured at the mid-axillary line (halfway between the lowest rib and the top of the iliac crest) and hip circumference was measured at the widest circumference of the hips using a tape measure. Blood pressure was measured once after at least 5 min of rest using automatic device (automatic blood pressure monitor, Tensoval comfort classic).

2.5. Resting metabolic rate

The resting metabolic rate was assessed with indirect calorimetry (Omnia 1.6.10. - Quark CPET, Cosmed) using the mask. Resting metabolic rate assessments were performed in the morning in a room with controlled temperature, sound and light, after a 10 min rest period of participant. Participants were measured in the supine position for 20 min. The first 5 min of data collection were excluded from the analysis.

2.6. Blood analysis

Venous blood samples were collected in 9 mL vacuum test tubes (Greiner Bio-One). Serum was immediately separated by a 10 min centrifugation at 2000×g, frozen, and stored at −80 °C until subsequent analysis. Serum concentrations of fasting glucose, triacylglycerols, total cholesterol, low-density lipoprotein (LDL) cholesterol, HDL cholesterol, CRP were measured with a cobas® c 111 analyzer.

2.7. Adherence measures

Compliance with the protocol was assessed using a daily dietary calendar. The dietary calendar documented adherence to the time window for food intake as well as any deviations from the energy

intake and number of meals that were not in accordance with the prescribed dietary plan. The participants were allowed to change their daily time window by 1 h. However, the time of the last meal had to be adjusted to the time of the first meal. Participants who did not adhere to the prescribed eating window or ER for four consecutive days during the intervention were excluded from the study. Moreover, at two months after the beginning of intervention eating plan adherence was assessed using a 3-day food record. Participants were instructed to record all foods and drinks consumed on two working days and one day off or weekend day in as much detail as possible. If this was not possible, they were asked to photograph their meals. Dietary data were analysed using the Open Platform for Clinical Nutrition (OPEN), which is accessible via the website <http://opkp.si/> (accessed on 1st March 2024). Moreover, diet quality was evaluated by calculating the Healthy Eating Index 2015 (HEI) as previously published [27].

2.8. Quality of life and sleep quality

At the beginning and the end of the study, participants completed the Pittsburgh Sleep Quality Index (PSQI) [28] and the WHO-5 Well-being Index (WBI) [29]. The PSQI evaluates seven components of sleep quality in the last 30 days: subjective sleep quality, sleep latency, sleep duration, sleep efficiency, sleep disturbances, use of sleeping medication, and daytime dysfunction. The score for each component can range from 0 (no difficulties) to 3 (severe difficulties). Summing the average scores of the seven factors gives a global PSQI score from 0 to 21, with 0–4 indicating “good” sleep and 5–21 indicating “poor” sleep [28].

The WHO-5 WBI is a short, self-administered, and positively worded scale designed to measure the level of subjective well-being over the last two weeks. The raw score is calculated as the sum of five items and ranges from 0 to 25, with 0 being the worst possible quality of life and 25 being the best possible quality of life. To obtain a percentage value between 0 and 100, the raw score is multiplied by 4 [29].

2.9. Statistical analysis

The sample size of the present study was calculated using the F-test and the ANOVA statistical test with repeated measures between factors in the statistical G x Power version 3.1 software. Body mass was specified as the primary outcome and was used for the power analysis. For the calculation, we used *per-protocol* data on body mass change from a previous study [30]. The result of the calculation was an effect size (f Cohen) of 0.30. For our primary outcome, we calculated that a total sample size of 75 participants would provide a power β of 80 % and a significance level (α) of 0.05, allowing an attrition rate of 20 %.

Statistical analysis was performed using SPSS version 29.0 (IBM Corp., Armonk, NY, USA). All analysis were *per-protocol*. The normality of variables were tested using the Shapiro–Wilk test. Data are presented as mean $\pm$ SD, mean (95 % confidence intervals (CI)). Categorical variables, such as sex, education, family status, place of residence, and working status are presented as frequencies and were analyzed using a chi-square test. The effects of the interventions within each group were analysed with a student's paired-samples t-test or Wilcoxon signed-rank test. The Kruskal–Wallis test or the one-way ANOVA test using Bonferroni post hoc test were used to evaluate differences in baseline characteristics between participants in all three groups and comparing the changes from baseline between groups in outcomes which were collected at baseline and at month 3. For those outcomes which were collected at 1, 2, and 3 months, a linear mixed model was applied, testing for the effect of time and the effect of

intervention with participants as a random intercept and with adjustment for age, sex, and BMI. The Bonferroni post hoc test was used to detect significant differences identified by linear mixed model. A significance level of 5 % ($p < 0.05$) was defined for all statistical tests.

3. Results

3.1. Adherence to intervention

A total of 300 individuals expressed an interest in the study and were screened based on completed questionnaire, inclusion and exclusion criteria. Further, 108 participants (75 females and 33 males) (Fig. 2) met the inclusion criteria and were invited to attend a first visit and to take initial measurements (body weight, body height, waist circumference, blood pressure, body composition analysis, blood collection, resting metabolic rate) at the faculty. Based on the MEQ, participants were allocated into three groups: eTRE + ER ($n = 37$), ITRE + ER ($n = 37$) and ER ($n = 34$). At the end of the 3-month intervention study, the number of drop outs were as follows: eTRE + ER group $n = 3$ participants (8 %), ITRE + ER group $n = 9$ participants (24 %), and ER group $n = 3$ participants (9 %). Reasons for drop out were personal reasons, scheduling conflicts, injuries during the intervention, difficulty following the eating schedule, medications, unknown reasons, and COVID 19 positive (Fig. 2). Therefore, 92 % (34 of 37) of the eTRE + ER, 76 % (28 of 37) of the ITRE + ER and 91 % (31 of 34) of the ER group were consistently adherent and completed the 3-month intervention study. For those who were adherent in the eTRE + ER group, ITRE + ER group and ER group the average eating window throughout the study was 8.1h (95 % CI, 7.8, 8.3), 8h (95 % CI, 7.9, 8.1) and 12.0 h (95 % CI, 11.9, 12.1), respectively. As expected, the eating window throughout the study decreased significantly in the eTRE + ER group and in the ITRE + ER

group compared to the pre-intervention period ($p = 0.001$) and compared to the ER group ($p = 0.001$) in which the eating window remained almost unchanged after 3 months of the intervention (Table 1). No serious adverse events were reported.

3.2. Allocation and baseline participants characteristics

Table 1 shows the baseline characteristics of all adherent participants allocated to the three different groups. At baseline, there were no statistically significant differences among the three groups in socio-demographic characteristics, body mass, body composition (BMI, % of body fat, visceral fat rating fat free mass, muscle mass, waist circumference, and metabolic age), physical activity, resting metabolic rate, blood pressure, energy intake or the three major macronutrients, HEI, PSQI, WHO-5 WBI, and in some cardiometabolic risk factors (fasting glucose, triacylglycerols, HDL cholesterol and CRP) (all p s > 0.05). However, a statistically significant difference between eTRE + ER and ITRE + ER group was found in total cholesterol and LDL cholesterol ($p = 0.037$, $p = 0.024$, respectively).

The analysis of 3-day food record showed that participants had good adherence to the prescribed eating plan with regard to energy, carbohydrate (45–50 %), protein (15–20 %) and fat intake 30–35 %) throughout the study. Mean changes in energy intake and macronutrients, and differences among the groups are given in Table 2. The mean energy deficit in % was -27 % (95 % CI, -33 – -21 , $p < 0.001$) for the eTRE + ER group, -20 % (95 % CI, -28 , -12 , $p < 0.001$) for the ITRE + ER group and -27 % (95 % CI, -32 , -22 , $p < 0.001$) for the ER group, with no significant differences among groups ($p = 0.194$). Similarly, participants in all three groups significantly decreased intake of carbohydrates (g/day), proteins (g/day) and fats (g/day), with no significant differences among the groups ($p = 0.705$, $p = 0.384$, $p = 0.191$, respectively). On the other hand, the between-group analysis showed

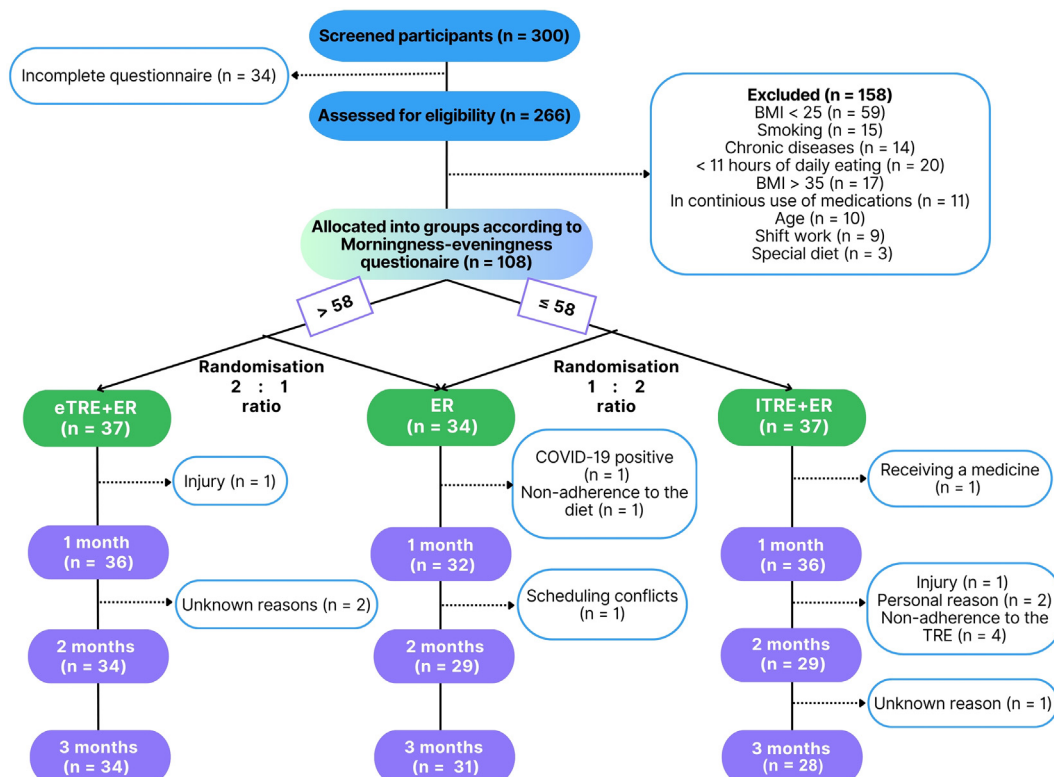


Fig. 2. Flowchart. BMI, body mass index; ER, energy restriction; eTRE, early time-restricted eating; ITRE, late time-restricted eating; n, total number.

Table 1
Baseline characteristics of study participants.

VARIABLES	eTRE + ER	ITRE + ER	ER	p value
	N (%) or M ± SD or 95 % CI	N (%) or M ± SD or 95 % CI	N (%) or M ± SD or 95 % CI	
N (F/M)	34 (26/8)	28 (18/10)	31 (19/12)	0.381
Socio-demographic determinants				
Age (years)	44 ± 9	46 ± 11	46 ± 9	0.568
Education				
Secondary school	5 (14.7 %)	6 (21.4 %)	10 (32.2 %)	0.788
Bachelor	20 (58.8 %)	16 (57.1 %)	16 (51.6 %)	
Master's degree	8 (23.5 %)	4 (14.3 %)	4 (12.9 %)	
Doctorate degree	1 (2.9 %)	2 (7.1 %)	1 (3.2 %)	
Family status				
Single	5 (14.7 %)	4 (14.3 %)	5 (16.1 %)	0.291
In a relationship	14 (41.2 %)	12 (42.9 %)	6 (19.4 %)	
Married	15 (44.1 %)	11 (39.3 %)	20 (64.5 %)	
Widowed	0 (0 %)	1 (3.6 %)	0 (0 %)	
Place of residence				
Village	13 (38.2 %)	3 (10.7 %)	8 (25.8 %)	0.205
Suburban area	8 (23.5 %)	10 (35.7 %)	10 (32.3 %)	
City	13 (38.2 %)	15 (53.6 %)	13 (41.9 %)	
Working status				
Employed	33 (97.1 %)	25 (89.2 %)	30 (100 %)	0.265
Unemployed	0 (0 %)	1 (3.6 %)	0 (0 %)	
Student	1 (2.9 %)	2 (7.1 %)	0 (0 %)	
Anthropometric parameters				
Body height (cm)	170.4 ± 9.7	173.3 ± 8.8	172.0 ± 8.5	0.399
Body mass (kg)	84.3 ± 12.5	86.8 ± 9.3	88.2 ± 16.0	0.481
BMI (kg/m ²)	28.9 ± 3.1	29.0 ± 2.1	29.6 ± 3.6	0.603
Waist circumference (cm)	94.7 ± 9.5	98.0 ± 6.9	97.6 ± 13.5	0.386
Fat mass (%)	33.9 ± 6.4	32.8 ± 6.8	32.6 ± 6.9	0.671
Visceral fat rating	8.1 ± 2.7	8.9 ± 2.3	9.5 ± 4.0	0.195
Fat free mass (kg)	55.8 ± 11.1	58.7 ± 10.6	59.4 ± 11.9	0.407
Muscle mass (kg)	53.0 ± 10.6	55.7 ± 10.2	56.4 ± 11.4	0.406
Metabolic age (years)	52 ± 11	52 ± 10	53 ± 11	0.865
Blood pressure				
Systolic blood pressure (120–140 mmHg)	134 ± 16	130 ± 11	128 ± 15	0.362
Diastolic blood pressure (80–90 mmHg)	88 ± 9	85 ± 9	83 ± 10	0.074
Cardiometabolic risk factors				
Fasting glucose (3.0–6.1 mmol/L)	5.14 ± 0.56	5.35 ± 0.48	5.17 ± 0.74	0.357
Total cholesterol (4.0–5.2 mmol/L)	5.08 ± 1.01	5.74 ± 1.12	5.19 ± 0.86	0.031^a
LDL cholesterol (2.0–3.3 mmol/L)	3.60 ± 0.98	4.33 ± 1.25	3.74 ± 0.92	0.022^a
HDL cholesterol (>1.45 mmol/L for males >1.68 mmol/L for females)	1.59 ± 0.44	1.55 ± 0.37	1.57 ± 0.43	0.933
Triacylglycerols (0.6–1.7 mmol/L)	1.34 ± 1.43	1.62 ± 1.55	1.40 ± 0.64	0.662
C-reactive protein (0.0–5.0 mg/L)	1.43 ± 0.77	1.81 ± 1.03	1.61 ± 1.31	0.503
Resting metabolic rate (kJ)	6545 ± 1520	7119 ± 1394	6796 ± 1491	0.334
Lifestyle characteristic				
Level of physical activity				
Low	3 (8.8 %)	4 (14.3 %)	9 (30.0 %)	0.161
Moderate	19 (55.9 %)	13 (46.4 %)	15 (50.0 %)	
High	12 (35.3 %)	11 (39.3 %)	6 (20.0 %)	
PSQI				
Subjective sleep quality	0.97 ± 0.80	0.86 ± 0.71	1.03 ± 0.66	0.647
Sleep latency	1.15 ± 0.99	1.25 ± 0.89	1.23 ± 0.85	0.895
Sleep duration	0.88 ± 0.60	0.79 ± 0.74	0.87 ± 0.56	0.813
Sleep efficiency	0.59 ± 0.96	0.61 ± 0.88	0.19 ± 0.48	0.078
Sleep disturbance	1.26 ± 0.45	1.14 ± 0.36	1.23 ± 0.43	0.510
Sleep medications	0.03 ± 0.17	0.11 ± 0.57	0.13 ± 0.56	0.658
Daytime dysfunction	0.85 ± 0.61	0.68 ± 0.55	0.97 ± 0.71	0.212
Global score	5.76 ± 2.98	5.43 ± 2.99	5.65 ± 2.37	0.894
WHO-5-WBI				
Feel cheerful and in a good mood	3.01 ± 0.93	3.46 ± 0.88	3.35 ± 0.84	0.228
Feel calm and relaxed	2.88 ± 1.01	3.14 ± 0.93	3.03 ± 0.88	0.567
Feel lively and full of energy	2.38 ± 1.05	2.79 ± 1.10	2.61 ± 0.96	0.308
Wake up fresh and rested	2.50 ± 1.35	2.75 ± 1.08	2.71 ± 1.04	0.660
My daily life filled with things that interest me	2.68 ± 1.12	3.14 ± 1.21	3.23 ± 0.92	0.095
Total	13.53 ± 4.20	15.29 ± 3.78	14.94 ± 3.46	0.160
Total score	54.1 ± 16.8	61.1 ± 15.1	59.7 ± 13.9	0.160
Food intake				
Energy intake (kJ)	9765 ± 1982	9479 ± 3570	9689 ± 2369	0.913
Carbohydrates (g/day)	260 ± 61	248 ± 96	253 ± 61	0.808
Carbohydrates (%)	45 ± 7	44 ± 5	44 ± 7	0.199
Protein (g/day)	96 ± 22	95 ± 41	92 ± 28	0.890
Protein (%)	17 ± 3	17 ± 3	16 ± 3	0.687
Fats (g/day)	95 ± 26	93 ± 38	98 ± 35	0.820
Fats (%)	36 ± 5	37 ± 4	38 ± 6	0.458

Table 1 (continued)

VARIABLES	eTRE + ER	ITRE + ER	ER	p value
	N (%) or M ± SD or 95 % CI	N (%) or M ± SD or 95 % CI	N (%) or M ± SD or 95 % CI	
Healthy eating index	59 ± 11	59 ± 14	61 ± 12	0.956
Eating window (h)				
Pre-intervention	12.9 (12.5, 13.2)	12.5 (12, 12.9)	12.9 (12.5, 13.3)	0.269
During intervention	8.1 (7.8, 8.3)	8 (7.9, 8.1)	12.0 (11.9, 12.1)	0.001 ^{b,c}

Abbreviations: BMI, body mass index; CI, confidence interval; ER, energy restriction; eTRE, early time restricted eating; F, female; ITRE, late time restricted eating; kg, kilograms; M, male; M, mean; SD, standard deviation; N, total number. p value denotes significant ($p < 0.05$) difference between eTRE + ER, ITRE + ER and ER group using ANOVA and the Bonferroni post hoc test for normal distribution, Kruskal–Wallis test for abnormal distribution and a chi-square test for categorical values.

p^a – denotes significant ($p < 0.05$) difference between eTRE + ER and ITRE + ER groups.

p^b – denotes significant ($p < 0.05$) difference between eTRE + ER and ER groups.

p^c – denotes significant ($p < 0.05$) difference between ITRE + ER and ER group.

Table 2

Adherence to intervention.

Variables	Participant group, mean change from baseline (95 % CI) ^a			Between-group comparison ^b			
	eTRE + ER	ITRE + ER	ER	p value	eTRE + ER vs. ITRE + ER	eTRE + ER vs. ER	ITRE + ER vs. ER
Food intake							
Energy intake (kJ)	–2824 (–3615, –2109)***	–2138 (–3142, –1134)***	–2732 (–3326, –2142)***	0.400	0.205	0.863	0.281
Energy intake (%)	–27 (–33, –21)***	–20 (–28, –12)***	–27 (–32, –22)***	0.194	0.111	0.986	0.115
Carbohydrates (g/day)	–75 (–97, –53)***	–62 (–90, –35)***	–72 (–92, –53)***	0.705	0.424	0.872	0.528
Carbohydrates (%)	–0.2 (–2.7, 2.3)	–1.1 (–4.5, 2.2)	–0.3 (–2.8, 2.1)	0.867	0.615	0.938	0.676
Protein (g/day)	–12 (–21, –4)*	–10 (–23, 3)	–3 (–12, 5)	0.384	0.705	0.177	0.357
Protein (%)	3.6 (2.5, 4.8)***	2.8 (1.3, 4.4)***	5.4 (4.1, 6.7)***	0.023	0.396	0.052	0.008
Fats (g/day)	–33 (–42, –24)***	–22 (–37, –8)**	–36 (–46, –27)***	0.191	0.172	0.664	0.081
Fats (%)	–2.6 (–4.9, –0.3)*	–1.2 (–4.7, 2.3)	–4.5 (–7.1, –1.9)*	0.249	0.468	0.318	0.099
Healthy eating index	19 (15, 23)***	16 (10, 23)***	20 (15, 26)***	0.607	0.514	0.715	0.325

Abbreviations: CI, confidence interval; ER, energy restriction; eTRE, early time restricted eating; ITRE, late time restricted eating.

^a Within-group evaluations of changes in outcome variables were performed with paired t test for normally distributed variables or paired Wilcoxon tests for abnormally distributed variables. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

^b Between-group comparisons were conducted using one-way ANOVA and the Bonferroni post hoc test for normally distributed variables or Kruskal–Wallis H tests for abnormally distributed variables.

Table 3

Effects of eTRE + ER, ITRE + ER and ER on anthropometric parameters, blood pressure and cardiometabolic risk factors.

Variables	Mean change from baseline (95 % CI)			Mean difference between groups (95 % CI)		
	eTRE + ER	ITRE + ER	ER	eTRE + ER vs. ITRE + ER	eTRE + ER vs. ER	ITRE + ER vs. ER
Anthropometric parameters						
Body mass (kg)	–5.0 (–5.7, –4.3)	–4.4 (–5.2, –3.6)	–4.3 (–5.0, –3.6)	–0.6 (–1.9, 0.7)	–0.7 (–2.0, 0.5)	–0.1 (–1.4, 1.2)
BMI (kg/m ²)	–1.7 (–2.0, –1.5)	–1.5 (–1.7, –1.2)	–1.4 (–1.7, –1.2)	–0.3 (–0.7, 0.2)	–0.3 (–0.7, 0.1)	–0.1 (–2.5, 1.2)
Fat mass (%)	–2.7 (–3.3, –2.2)	–1.6 (–2.1, –1.0)	–1.7 (–2.2, –1.1)	–1.2 (–2.1, –0.2)	–1.1 (–2.0, –0.1)	–0.1 (–0.9, –1.1)
Visceral fat rating	–1.1 (–1.3, –0.9)	–0.8 (–1.0, –0.5)	–0.9 (–1.1, –0.6)	–0.3 (–0.7, 0.1)	–0.2 (–0.7, 0.2)	0.1 (–0.4, 0.5)
Fat free mass (kg)	–1.2 (–1.6, –0.9)	–1.7 (–2.1, –1.3)	–1.5 (–1.9, –1.2)	–0.5 (–0.2, 1.2)	0.3 (–0.3, 1.0)	–0.2 (–0.8, 0.5)
Muscle mass (kg)	–1.2 (–1.5, –0.8)	–1.6 (–2.0, –1.2)	–1.5 (–1.8, –1.1)	0.5 (–0.2, 1.1)	0.3 (–0.3, 0.9)	–0.1 (–0.8, 0.5)
Waist circumference (cm)	–3.5 (–4.1, –2.8)	–3.1 (–3.8, –2.4)	–2.9 (–3.5, –2.3)	–0.4 (–1.5, 0.8)	–0.5 (–1.6, 0.5)	–0.2 (–1.3, 1.0)
Metabolic age (years)	–6 (–7, –5)	–4 (–5, –2)	–3 (–5, –2)	–2 (–5, 0)	–3 (–5, –0)	–1 (–3, 2)
Resting metabolic rate (kJ)	–549 (–948, –194)	–459 (–746, –173)	–238 (–631, 274)	–89 (–775, 596)	–311 (–978, 355)	–221 (–917, 474)
Blood pressure						
Systolic blood pressure (mmHg)	–8 (–11, –4)	–7 (–11, –3)	–7 (–10, –3)	–1 (–7, 6)	–1 (–7, 5)	–1 (–7, 6)
Diastolic blood pressure (mmHg)	–10 (–12, –8)	–7 (–10, –5)	–6 (–8, –3)	–3 (–7, 1)	–4 (–8, –0)	–1 (–6, 3)
Cardiometabolic risk factors						
Fasting glucose (mmol/L)	–0.48 (–0.63, –0.32)	–0.13 (–0.30, 0.05)	–0.20 (–0.37, –0.04)	–0.35 (–0.63, –0.06)	–0.27 (–0.55, 0.01)	0.08 (–0.22, 0.37)
Total cholesterol (mmol/L)	–0.30 (–0.46, –0.14)	–0.12 (–0.31, 0.06)	–0.27 (–0.44, –0.10)	–0.18 (–0.47, 0.12)	–0.03 (–0.32, 0.26)	0.15 (–0.16, 0.45)
LDL cholesterol (mmol/L)	–0.23 (–0.38, –0.07)	–0.01 (–0.18, 0.17)	–0.23 (–0.39, –0.06)	–0.22 (–0.51, 0.07)	–0.00 (–0.28, 0.28)	0.22 (–0.08, 0.51)
HDL cholesterol (mmol/L)	–0.06 (–0.12, –0.01)	–0.06 (–0.12, –0.00)	–0.04 (–0.09, 0.02)	–0.00 (–0.10, 0.09)	–0.03 (–0.12, 0.06)	–0.03 (–0.12, 0.07)
Triacylglycerols (mmol/L)	–0.18 (–0.32, –0.04)	–0.24 (–0.39, –0.09)	–0.21 (–0.36, –0.07)	0.06 (–0.19, 0.31)	–0.04 (–0.21, 0.28)	–0.02 (–0.28, 0.23)
C-reactive protein (mg/L)	–0.28 (–0.66, 0.11)	–0.49 (–0.88, –0.11)	0.00 (–0.38, 0.38)	0.22 (–0.46, 0.89)	–0.28 (–0.94, 0.39)	–0.50 (–1.16, 0.17)

Abbreviations: BMI, body mass index; CI, confidence interval; ER, energy restriction; eTRE, early time-restricted eating; ITRE, late time-restricted eating. Means were estimated using a *per protocol* analysis using linear mixed model and the Bonferroni post hoc test to detect significant differences between groups. Errors bars indicate 95 % CI for each parameter from baseline by group.

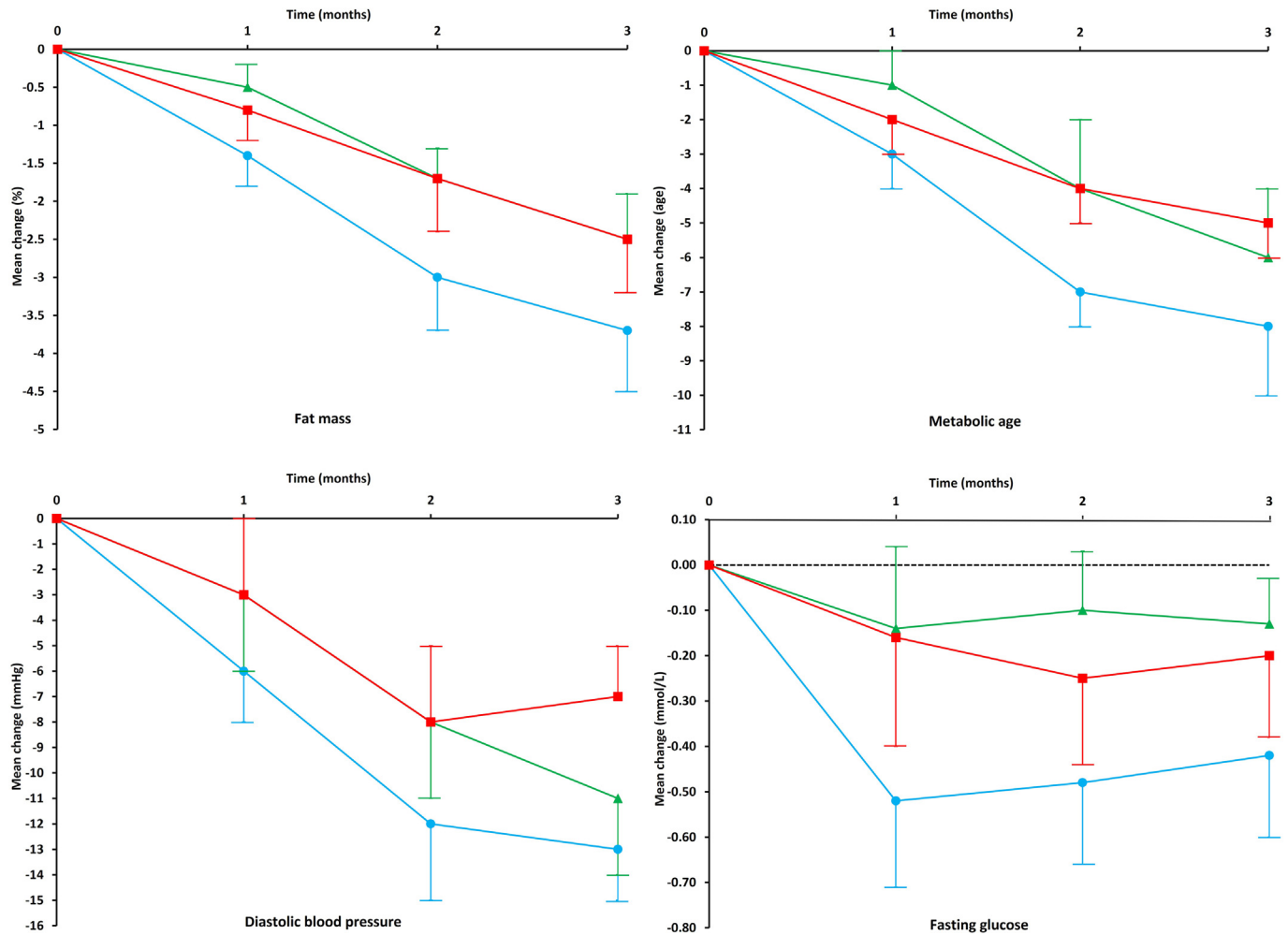


Fig. 3. Mean change in fat mass (%), metabolic age, diastolic blood pressure and fasting glucose in the study groups. Data are visualized as mean (95 % CI); ER, energy restriction; eTRE, early time-restricted eating; ITRE, late time-restricted eating.

a significant decrease in the proportion of protein to energy intake in the ITRE + ER group compared to the ER group ($p = 0.008$). However, HEI significantly increased in all three groups (all p s < 0.001), with no significant differences among them ($p = 0.607$).

3.3. Effects of eTRE + ER, ITRE + ER and ER intervention on anthropometric parameters

Changes in body mass and body composition are reported in Table 3. There was a significant time main effect ($p < 0.001$), suggesting a decrease in body mass at the end of 3-month intervention with a mean loss of -5.0 kg (95 % CI, -5.7 , -4.3) for eTRE + ER group, -4.4 kg (95 % CI, -5.2 , -3.6) for ITRE + ER and -4.3 kg (95 % CI, -5.0 , -3.6) for the ER group, with no significant difference among groups ($p = 0.319$). Total body weight loss was -6 %, -5 % and -5 % for the eTRE + ER group, ITRE + ER group, and ER group, respectively. Likewise, a significant decrease was also shown in BMI, visceral fat rating, fat free mass, muscle mass, waist circumference, and resting metabolic rate in all three groups, without a significant difference between them. On the other hand, after 3-month of the intervention there was a significant difference in % of fat mass between the eTRE + ER and ITRE + ER group (-1.2 % (95 % CI, -2.1 , -0.2 , $p = 0.013$), and between eTRE + ER and ER group (-1.1 % (95 % CI, -2.0 , -0.1 , $p = 0.022$) and in metabolic age

between eTRE + ER and ER group (-3 years (95 % CI, -5 , -0 , $p = 0.028$) (Fig. 3).

3.4. Effects of eTRE + ER, ITRE + ER and ER intervention on blood pressure and cardiometabolic risk factors

Furthermore, we compared the effects of eTRE + ER group, ITRE + ER group and ER group on blood pressure and cardiometabolic risk factors (Table 3).

There was a significant time main effect ($p < 0.001$), suggesting a decrease in diastolic blood pressure, systolic blood pressure, LDL cholesterol, and total cholesterol, a significant time main effect ($p < 0.01$), suggesting a decrease in HDL cholesterol and a significant time main effect ($p < 0.05$), suggesting a decrease in CRP at the end of the 3-month intervention. After 3 months of the intervention there was a significant difference in fasting glucose between eTRE + ER group and ITRE + ER group, where eTRE + ER group lowered fasting glucose by an additional 0.35 mmol/L (95 % CI, -0.63 , -0.06 , $p = 0.012$) relative to the ITRE + ER group (Fig. 3). Moreover, the eTRE + ER group lowered diastolic blood pressure by an additional 4 mmHg (95 % CI, -8 , -0 , $p = 0.033$) compared to the ER group (Fig. 3).

No statistically significant differences in total cholesterol ($p = 0.322$), LDL cholesterol ($p = 0.161$), HDL cholesterol ($p = 0.701$), triacylglycerols ($p = 0.837$), CRP ($p = 0.196$), and in

Table 4
Effects of eTRE + ER, ITRE + ER and ER on sleep quality and well-being.

Variables	Participant group, mean change from baseline (95 % CI) ^a			Between-group comparison ^b			
	eTRE + ER	ITRE + ER	ER	p value	eTRE + ER vs. ITRE + ER	eTRE + ER vs. ER	ITRE + ER vs. ER
PSQI							
Sleep quality	−0.21 (−0.46, 0.05)	−0.21 (−0.48, 0.05)	−0.19 (−0.43, 0.05)	0.511	0.643	0.468	0.252
Sleep latency	−0.29 (−0.57, −0.02)	−0.29 (−0.52, 0.05)	−0.10 (−0.42, 0.22)	0.754	0.495	0.551	0.921
Sleep duration	0.03 (−0.17, 0.23)	0.25 (0.02, 0.48)	0.13 (−0.08, 0.34)	0.623	0.337	0.958	0.415
Sleep efficiency	−0.29 (−0.60, −0.01)	0.07 (−0.26, 0.40)	0.23 (−0.08, 0.54)	0.047	0.756	0.017	0.492
Sleep disturbance	−0.06 (−0.21, 0.09)	−0.11 (−0.30, 0.09)	−0.23 (−0.43, −0.02)	0.385	0.703	0.177	0.360
Sleep medications	−0.00 (−0.00, 0.00)	−0.11 (−0.33, 0.11)	−0.00 (−0.28, 0.28)	0.686	0.443	0.998	0.452
Daytime dysfunction	−0.06 (−0.30, 0.18)	−0.07 (−0.33, 0.19)	−0.19 (−0.43, 0.05)	0.682	0.942	0.421	0.487
Global score	−0.85 (−1.60, −0.11)*	−0.46 (−1.29, 0.36)	−0.36 (−1.23, 0.52)	0.637	0.495	0.369	0.851
WHO-5 WBI							
Feel cheerful and in a good mood	0.41 (0.08, 0.74)*	0.29 (−0.08, 0.65)*	0.35 (0.06, 0.65)*	0.861	0.585	0.800	0.769
Feel calm and relaxed	0.44 (0.07, 0.81)*	0.50 (0.14, 0.86)*	0.42 (0.11, 0.73)*	0.945	0.809	0.926	0.745
Feel lively and full of energy	0.76 (0.40, 1.13)***	0.79 (0.36, 1.21)***	0.74 (0.46, 1.02)***	0.985	0.933	0.926	0.865
Wake up fresh and rested	0.76 (0.33, 1.20)**	0.54 (0.19, 0.88)**	0.58 (0.23, 0.93)**	0.658	0.398	0.485	0.871
My daily life filled with things that interest me	0.44 (0.14, 0.74)	0.32 (−0.10, 0.74)	0.06 (−0.39, 0.52)	0.357	0.661	0.158	0.358
Total	2.88 (1.43, 4.33)***	2.43 (0.93, 3.93)**	2.16 (0.99, 3.34)***	0.739	0.639	0.444	0.787
Total Score	11.53 (5.73, 17.33)**	9.71 (3.71, 15.72)**	7.19 (0.86, 13.53)*	0.572	0.668	0.293	0.560

Abbreviations: CI, confidence interval; ER, energy restriction; eTRE, early time restricted eating; ITRE, late time restricted eating, PSQI, Pittsburgh sleep quality index; WHO-5 WBI, World Health Organization Well-being index.

^a Within-group evaluations of changes in outcome variables were performed with paired t test for normally distributed variables or paired Wilcoxon tests for abnormally distributed variables. *p < 0.05, **p < 0.01, ***p < 0.001.

^b Between-group comparisons were conducted using one-way ANOVA and the Bonferroni post hoc test for normally distributed variables or Kruskal–Wallis H tests for abnormally distributed variables.

systolic blood pressure (p = 0.927) were found between the groups after 3 months of the intervention.

3.5. Effects of eTRE + ER, ITRE + ER and ER intervention on sleep quality and quality of life

Table 4 shows the effects of eTRE + ER, ITRE + ER and ER on changes in sleep quality and general well-being. In all three groups global PSQI score significantly decreased after 3 months of intervention as compared to their baseline values. Similar was shown for almost all PSQI constructs (sleep quality, sleep latency, sleep disturbance, sleep medication and daytime symptoms), except for sleep efficiency, where there was a statistically significant mean difference between the eTRE + ER and the ER group (p = 0.017).

Moreover, Table 4 shows the mean changes in the total score for the WHO-5 WBI questionnaire and the scores for the individual items. Participants in all three groups achieved significantly higher total score compared to baseline, and no significant differences between the groups were observed.

4. Discussion

This study was conducted to investigate the effects of eTRE + ER and ITRE + ER vs. ER on body mass, body composition and cardiometabolic risk factors, sleep quality and quality of life in participants with overweight and obesity. As we were also interested in the mechanism of TRE and not just the metabolic health benefits, *per-protocol* analyses were conducted to analyse participants adhering to and completing the 3-month intervention.

The main findings of our study are that eTRE + ER, ITRE + ER and ER may provide similar benefits for body mass, body composition and metabolic parameters. However, eTRE + ER showed greater improvements in % fat mass and fasting glucose compared to the ITRE + ER group. In addition, eTRE + ER showed greater benefits in % fat mass, metabolic age and diastolic blood pressure compared to the ER group after 3 months of intervention.

TRE has been shown to often lead to an unintended reduction in energy intake under *ad libitum* conditions [9,31,32]. Therefore, in recent years, research has focused on the benefits of TRE in conjunction with ER to further clarify the effects of TRE + ER on weight loss and cardiometabolic risk. Moreover, an emerging body of evidence suggests that the timing of food intake in relation to personal chronotype may be involved in modulating metabolic processes [33,34]. Therefore, the results of TRE in humans seem to depend on the time of day of the eating window. However, it is still controversial which choice of eating time has better effects on weight loss, body composition and cardiometabolic health [35–37]. Our study showed that eTRE + ER provided a greater benefit in decreasing the percentage of fat mass compared with the ITRE + ER and ER groups, and a greater benefit in decreasing metabolic age compared with the ER group. No other differences in the benefits of eTRE + ER, ITRE + ER or ER were found in weight loss, BMI, waist circumference or other anthropometric data in our study, at least not when participants were prescribed the same dietary plan. In agreement with our study, one study also found that participants who adhered to eTRE + ER for more than 5 days/week every week had greater improvements in fat mass compared to the control group [38]. On the other hand, the same study found additional benefits of eTRE + ER compared to our study, such as a greater

improvement in body mass. The same was also found in two recent meta-analyses, in which eTRE + ER significantly reduced body mass and fat mass compared to ER alone [39] and eTRE + ER showed a greater effect on body mass compared to lTRE + ER [40]. In addition, another study found that eTRE + ER was more effective in weight loss but had no effect on body fat [41]. In contrast, another study reported no differences in the benefits of eTRE + ER, lTRE + ER and ER on weight loss and body composition in overweight or obese individuals [30].

One mechanism by which eTRE could affect adipose tissue and increase fat mass loss more efficiently than lTRE + ER and ER alone could be by increasing lipolysis and beta-oxidation [42]. In the morning, during the extended fasting period after a meal, glycogen stores in the body are reduced, which leads to an increased utilization of fats as the primary source of energy [43]. Moreover, it has been shown that fatty acid oxidation in skeletal muscles and the thermic effect of food are also higher in the morning [44]. In addition, a regulated circadian clock regulates the secretion of several hormones that are important for metabolic regulation and reach their maximum secretion in the morning during the active phase. For example, adiponectin, which stimulates fatty acid oxidation and glycolysis and inhibits fat accumulation, reaches its peak secretion at 11:00 [45]. In addition, another hormone, fibroblast growth factor (FGF)-21, is secreted by the liver between 4:00 and 9:00, which promotes fat oxidation [46]. Thus (FGF)-21 and adiponectin peak in secretion in the morning, which promote fatty acid oxidation and glycolysis and inhibit fat accumulation [46,47]. Another possibility is that the thermic effect of food is greater in the morning due to the circadian system [48,49]. Studies have shown that the insulin response to a meal is more efficient in the morning, while the reduced thermic response in the evening may be a result of nocturnal insulin resistance [50,51]. Glucose tolerance worsens in the evening and reaches its minimum around midnight. This daily variation in the thermic effect of food contributes significantly to the difference in energy balance, with energy intake being higher in the evening than in the morning [19,49,51].

Very little is also known about the effects of TRE + ER vs. ER on metabolic age. To our knowledge, our study is the only study comparing the effects of eTRE + ER and lTRE + ER vs. ER on metabolic age. Reduced metabolic age in eTRE + ER was most likely due to a significantly reduced body fat mass (%) [52]. However, we have to point out that although metabolic age is not a clinically validated parameter, some studies have recently hypothesised that could serve as a predictive factor for metabolic syndrome [53,54]. In addition, the European Society of Cardiology has used metabolic age as one of the predictors of cardiovascular disease in its studies in individuals whose metabolic age was higher than their chronological age [55].

Regarding blood pressure, the results remain controversial. In agreement with our results, one study has demonstrated a beneficial effect of eTRE + ER on diastolic blood pressure [41]. Moreover, 5 weeks of eTRE in combination with prescribed dietary intervention showed surprisingly large improvement in morning levels of diastolic blood pressure and systolic blood pressure, relative to the 12 h control eating schedule [10]. On the other hand, eTRE + ER did not improve systolic blood pressure or diastolic blood pressure in the study using a *per-protocol* analysis [56]. One possible mechanism for lowering diastolic blood pressure could be an alignment of meal timing with the endogenous circadian rhythm when eating window is shifted to earlier in the day. Indeed, earlier in the day-time sodium excretion is upregulated by the circadian system and since eTRE shifts salt intake to early hours, natriuresis is promoted [56], which may benefit blood pressure. Moreover, previous study suggest that eTRE lowers insulin levels, which may be mechanistically associated with a decrease in blood pressure [10].

Glucose and lipid metabolism are regulated by the circadian system [57]. eTRE has been shown to result in improvements in glucose homeostasis [11,58]. Moreover, previous data suggest that eTRE may be more effective at reducing fasting glucose when compared to lTRE or ER alone and this was also the case in our study [11,17,59]. Studies on circadian rhythms have confirmed that the insulin sensitivity and β -cell function are better in the morning, so on this basis, eTRE could improve glucose metabolism [10,60,61]. Moreover, epidemiological studies in humans report the positive effect of morning energy intake on the prevention of metabolic disorders such as glucose intolerance and dyslipidemia [1].

Regarding the lipid profile, we did not find any significant between-groups differences in total cholesterol, triacylglycerols, HDL cholesterol and LDL cholesterol after 3 months of intervention. Our results are in agreement with majority of other previous studies showing that lipid profile was unaffected by the TRE protocol [9,30,32,58,62]. Moreover, a recently published meta-analysis also showed that neither eTRE + ER nor lTRE + ER vs. ER had a significant effect on lipid profiles [40,63].

Research on TRE, quality of life and sleep is very scarce. Some studies reported increased health-related quality of life and improved quality of sleep [15,16], whereas others reported that effects of TRE on sleep remain unclear [64,65]. In our study, we found an improvement in the global score of PSQI after 3 months of intervention in all three groups, but with no differences between them, which is similar as one previous study [30]. In our study the only significant difference among groups was in one construct of self-reported questionnaire - sleep efficiency - where participants who adhered to eTRE + ER reported less difficulties in sleep efficiency compared to participants in ER group. In contrast to our results, a previous study showed that participants who adhered to eTRE + ER ≥ 5 days/week every week sleeping less and taking longer to fall asleep compared to ER group [38]. However, the studies mainly used self-reported data and future studies with more objective measures are needed to clarify the effects of TRE on sleep quality.

General well-being in our study improved in all three groups, but with no significant difference among them. Previously, in a 5-week weight loss trial, individuals following an energy-restricted eTRE schedule reported having improved mood compared with the ER group [38]. In that study, individuals in the eTRE + ER group also had a significantly higher reduction in body mass compared with the ER group, which could have influenced the result. Indeed, it has been shown that increase in self-reported quality of life can be explained by weight loss [3,66].

The main strength of our study is that we did not only focus on the timing of eating but also on quantity and quality of food consumed. All groups consumed on average 25 % less of energy intake compared with baseline, mainly at the expense of a lower daily total fat intake. By having the same structured, nutritious and energy-balanced meals and ER in all three groups, we ensured that the eating window was the only difference among groups. Moreover, we allocated the participants as closely as possible into three different groups according to their personal chronotype, thus ensuring that the participants' eating time corresponds to their lifestyle, and that the eating time is in line with their circadian clock.

Some limitations of our study should be acknowledged. First, the intervention was short-term, the 3-month intervention limits the ability to assess the long-term sustainability and health benefits of the dietary strategies and further studies are needed to determine whether the TRE dietary strategy can be effective in the maintenance stage. Second, despite best efforts, measurements in the eTRE + ER group were taken after 14–15 h of fasting, whereas in the lTRE + ER and ER groups they were taken after 13–14 h of

fasting, which may have contributed to a bias in the results and have influenced biochemical outcomes. Third, our study sample was predominantly female (80 %), which might affect the generalizability of the results to a broader population. In addition, food intake was only recorded for 3 days before and during the study, which is relatively low for a detailed analysis of energy intake and macronutrients, and it is also known that overweight people often underreport food intake. As a final limitation, in a recent systematic review [67], individuals with the evening chronotype were associated with lower weight loss success, a tendency to have a higher BMI and long-term weight control than individuals with the morning chronotype. Our groups, ITRE + ER and ER, could therefore be less successful than eTRE + ER due to the higher proportion of participants with the evening chronotype. Nevertheless, we reduced the effect of only evening types in ITRE + ER by including intermediate types.

5. Conclusions

To conclude, a 3-month intervention with eTRE + ER, ITRE + ER and ER alone provided similar weight loss benefits when participants were given diet plan with three meals and same macronutrient contribution. In addition, eTRE under ER conditions showed greater benefits for fasting blood glucose and percentage of fat mass compared to ITRE + ER, and greater benefits for percentage of fat mass, metabolic age and diastolic blood pressure compared to ER alone. However, despite the promising results of our study, there is limited evidence directly comparing the effects of eTRE + ER with ITRE + ER. Most studies focus on the effects of eTRE + ER, while the effects of ITRE + ER in human health are much less researched. In the future, well-designed studies with homogeneous and large samples are needed to ensure scientifically objective and reliable study results and to determine the effectiveness of eTRE + ER and ITRE + ER compared to ER on health-related parameters and quality of life.

Author contributions

AP, ZJP, NM, SK, TČ, and BH designed research; TČ and BH conducted research; AP, ZJP, and TČ analyzed data or performed statistical analysis; TČ, ZJP and AP wrote the paper. AP had primary responsibility for final content. All authors read and approved the final manuscript.

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Conflict of interest

The authors declare that they have no conflict of interest.

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